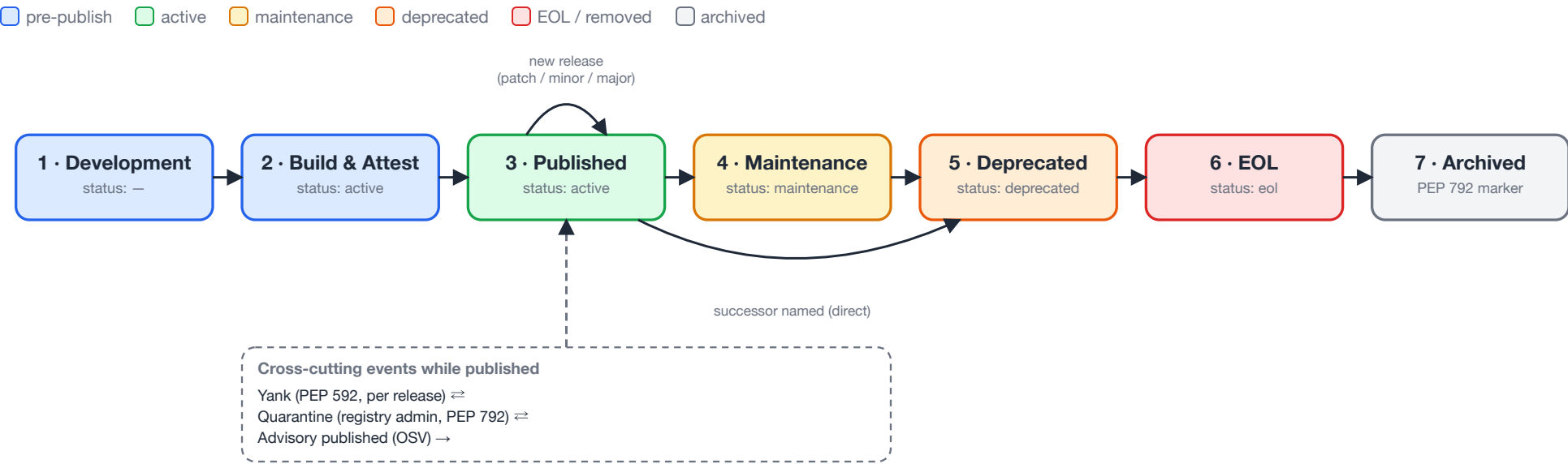


# Miri Artifact Lifecycle — Python Wheels

From first build to end-of-life: stages, metadata state, consumer signals, and the standards trail (PEP 702 · 770 · 592 · 792 · 387 · OSV · OpenEoX)

## 1 The Artifact Lifecycle



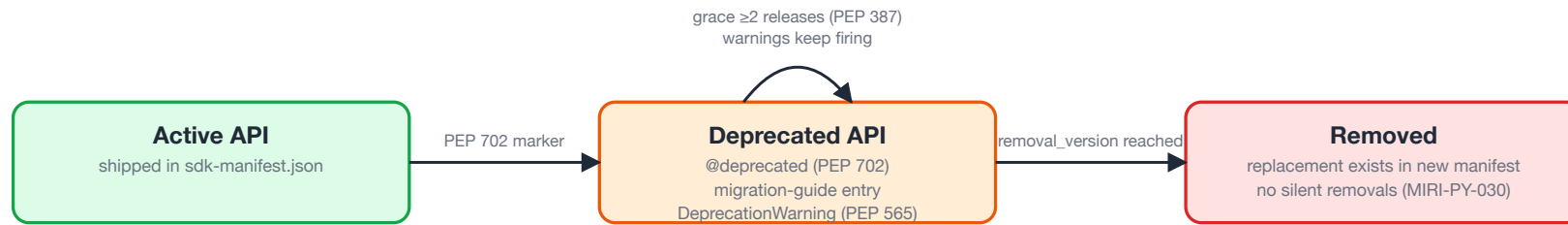
| Stage                  | support.status | Metadata actions                                                                                                                                                               | Consumers see                                                      | Standards                             |
|------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------|
| 1 · Development        | —              | Code carries @deprecated markers (PEP 702) as APIs evolve; metadata authored as data                                                                                           | Nothing — unpublished                                              | PEP 702                               |
| 2 · Build & Attest     | active         | Build generates agent-metadata/ (sdk-manifest, usage-patterns, migration-guide from PEP 702, lifecycle.json); PEP 770 SBOM if bundling non-Python; linter scores; attestations | Miri score gates release in CI                                     | Agent Metadata §5 · PEP 770 · PEP 740 |
| 3 · Published (Active) | active         | Registry serves; update_check answers latest; advisories attach via OSV; each release regenerates metadata + migration-guide                                                   | Full Miri surface: identity, advisory join, update state, examples | Lifecycle §3–4 · OSV                  |
| 4 · Maintenance        | maintenance    | Security fixes only; supported_versions narrows; deprecations accumulate                                                                                                       | Reduced-support signal before adoption                             | Lifecycle §3.1                        |
| 5 · Deprecated         | deprecated     | replacement purl set; final releases carry updated support; migration-guide documents the path out                                                                             | Every future install carries the successor pointer                 | Lifecycle §4.3                        |

| Stage               | support.status | Metadata actions                                                      | Consumers see                                                    | Standards                   |
|---------------------|----------------|-----------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------|
| <b>6 · EOL</b>      | eol            | eol_date passed; last release terminal; advisories may still be filed | Any use treated as a finding;<br>OpenEoX statement once ratified | Lifecycle §3.3 ·<br>OpenEoX |
| <b>7 · Archived</b> | eol            | PEP 792 archived marker at the registry; project read-only            | Index APIs signal archived state                                 | PEP 792                     |

## 2 Cross-Cutting Events

| Event                      | Mechanism                          | Effect                                                                | Standards            |
|----------------------------|------------------------------------|-----------------------------------------------------------------------|----------------------|
| <b>Advisory published</b>  | GHSA / PYSEC record → OSV          | advisory_sources join flags affected versions; fixed names the remedy | Lifecycle §4.2 · OSV |
| <b>Release yanked</b>      | PEP 592                            | Version leaves dependency resolution; pinned installs still work      | PEP 592              |
| <b>Project quarantined</b> | Registry administrators            | Entire project uninstallable until cleared                            | PEP 792              |
| <b>VEX statement</b>       | OpenVEX / CycloneDX VEX at vex URL | Affected-but-not-exploitable suppressions, auditable                  | OpenVEX              |

### 3 The Nested Interface Lifecycle (inside stages 3–5)



**Coherence rule:** Active → Removed without passing through Deprecated is a conformance failure (no silent removals — checklist MIRI-PY-030). Replacements must exist in the release that performs the removal; `removal_version` must exceed the current version.

### 4 The Three Clocks

Three versions advance independently; collapsing any two loses the ability to change one without a major bump in the other.

